A three-leaf counterexample for Fitch reconstruction under a molecular clock

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Abstract

Does Fitch parsimony on a tree whose leaves are all the same distance from the root reconstruct the root state at least as accurately as any single leaf? We give a negative answer for the symmetric twenty-state model. A rooted binary tree with three leaves and strictly positive edge lengths has Fitch accuracy 38829/400000, whereas each leaf agrees with the root with probability 39/400. The strict deficit is 171/400000. The proof is an exact calculation over four types of Fitch set and includes uniform final tie breaking.

1 Model and result

Ancestral-state reconstruction estimates a character at a common ancestor from its observed values at present-day leaves. Li, Steel and Zhang [1, p. 652] conjectured that, under a symmetric model on an ultrametric tree, Fitch parsimony using all leaves is at least as accurate as using any one leaf. Here ultrametric means that every root-to-leaf path has the same total length. The two-state case was proved by Fischer and Thatte [2, author version, Theorem 2], and the three- and four-state cases by Herbst and Fischer [3, author version, Theorems 1–2]. We exhibit a twenty-state counterexample. It negates the universal assertion over alphabet sizes $r \geq 5$, without asserting failure at each such size.

Let $r \geq 2$ be an integer. On the alphabet $\{1, \ldots, r\}$, choose the state at the root ρ uniformly, and evolve states independently down branches conditional on their parent states. Write σ_x for the state at a vertex x. An edge of length $\ell \geq 0$ uses the continuous-time Markov generator

$$Q_{ij} = \begin{cases} -1, & i = j, \\ 1/(r-1), & i \neq j. \end{cases} \tag{1}$$

Thus the expected substitution rate is one. If every leaf has height H, the probability that a specified leaf has the root state is

$$q_r(H) = \frac{1}{r} + \frac{r-1}{r} \exp\left(-\frac{rH}{r-1}\right). \tag{2}$$

The Fitch set at a leaf is its observed singleton. At each internal vertex, take the intersection of the two child sets when it is nonempty and their union otherwise. Choose the root estimate $\widehat{\sigma}_\rho$ uniformly from the resulting root set. Write $\mathrm{RA} = \mathbb{P}(\widehat{\sigma}_\rho = \sigma_\rho)$, including this final random choice. Only the topology and leaf states enter the set recursion.

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Theorem 1. *Let $r = 20$. Let the root ρ have children a, v, and let v have children b, c, so that the leaves are a, b, c. Give the edges the lengths*

$$\ell_{\rho a} = \frac{19}{20}\log 20, \qquad \ell_{\rho v} = \frac{19}{20}\log 16, \qquad \ell_{vb} = \ell_{vc} = \frac{19}{20}\log\frac{5}{4}. \tag{3}$$

All four lengths are finite and strictly positive, and every leaf has height $H = (19/20)\log 20$. Under (1) *and the uniform root and tie-breaking conventions above,*

$$\mathrm{RA} = \frac{38829}{400000} < \frac{39}{400} = q_{20}(H), \qquad q_{20}(H) - \mathrm{RA} = \frac{171}{400000}. \tag{4}$$

2 Exact calculation

Proof of Theorem 1. All logarithms are natural. The clock condition follows from $\log 16 + \log(5/4) = \log 20$. To obtain the edge probabilities with the stated rate, let Π be the matrix all of whose entries are $1/r$. Since $\Pi^2 = \Pi$ and $Q = -r(I-\Pi)/(r-1)$, the exponential series gives

$$e^{\ell Q} = \Pi + \lambda(I - \Pi), \qquad \lambda = \exp\left(-\frac{r\ell}{r-1}\right). \tag{5}$$

In particular, an edge preserves its input state with probability $(1+(r-1)\lambda)/r$ and changes it to each specified different state with probability $(1-\lambda)/r$.

Condition for now on $\sigma_\rho = 1$. The values of λ on the edges ρa, ρv, vb and vc are $1/20$, $1/16$, $4/5$ and $4/5$, respectively. Hence

$$\begin{aligned} q := \mathbb{P}(\sigma_a = 1 \mid \sigma_\rho = 1) = \frac{39}{400}, \quad \mathbb{P}(\sigma_a = j \mid \sigma_\rho = 1) = \frac{19}{400} =: p \quad (j \neq 1), \\ \mathbb{P}(\sigma_v = 1 \mid \sigma_\rho = 1) = \frac{7}{64}, \quad \mathbb{P}(\sigma_v \neq 1 \mid \sigma_\rho = 1) = \frac{57}{64}. \end{aligned} \tag{6}$$

Given any state at v, each of b, c equals that state with probability $u = 81/100$ and equals each specified different state with probability $w = 1/100$. The two terminal evolutions are independent conditional on σ_v.

Let B be the Fitch set at v. It is a singleton when $\sigma_b = \sigma_c$ and a two-element set otherwise. Its total singleton probability is $u^2 + 19w^2 = 329/500$, independent of the state at v. Denote the probabilities of the four possibilities by c_1, n_1, c_2, n_2, where c_k corresponds to $|B| = k$ and $1 \in B$, and n_k to $|B| = k$ and $1 \notin B$. All probabilities in the next calculation remain conditional on $\sigma_\rho = 1$. We have

$$\begin{aligned} c_1 &= \frac{7}{64}u^2 + \frac{57}{64}w^2 = \frac{1437}{20000}, & n_1 &= \frac{329}{500} - c_1 = \frac{11723}{20000}, \\ c_2 &= \frac{7}{64}\,2u(1-u) + \frac{57}{64}\,2w(1-w) = \frac{513}{10000}, & n_2 &= 1 - \frac{329}{500} - c_2 = \frac{2907}{10000}. \end{aligned} \tag{7}$$

Indeed, c_1 is the probability that both terminal states equal 1, and c_2 is the probability that exactly one of them equals 1. These four events are disjoint, exhaustive and have probabilities summing to one.

Conditional on the root, the leaf a is independent of the entire subtree below v, and therefore independent of B. Fix a possible set B of size $k \in \{1, 2\}$. If $1 \in B$, the value $\sigma_a = 1$ produces a correct singleton at the root. A value in $B \setminus \{1\}$ produces an incorrect singleton. Each of the $20 - k$ states outside B produces the union $B \cup \{\sigma_a\}$ of size $k + 1$, with success probability $1/(k+1)$. Thus the conditional success probability is

$$\begin{cases} q + (20-k)p/(k+1), & 1 \in B, \\ q/(k+1), & 1 \notin B. \end{cases} \tag{8}$$

For the second case, only $\sigma_a = 1$ can yield success, and it yields a union of size $k+1$. No state is randomly selected at v; the full set B is used in this merge.

Substituting $k = 1, 2$ into (8), then averaging the four events in (7), gives

$$\begin{aligned}\mathbb{P}(\widehat{\sigma}_\rho = 1 \mid \sigma_\rho = 1) &= c_1\frac{439}{800} + n_1\frac{39}{800} + c_2\frac{153}{400} + n_2\frac{13}{400} \\ &= \frac{630843 + 457197 + 313956 + 151164}{16000000} = \frac{38829}{400000}. \end{aligned} \tag{9}$$

Permuting all state labels preserves the transition probabilities, the Fitch set operations and the uniform final choice. The same conditional accuracy therefore holds for every root state, and averaging the uniform root law gives the asserted value of RA.

Finally, (2) and $\exp(-20H/19) = 1/20$ give $q_{20}(H) = 39/400 = 39000/400000$. Subtraction proves the strict deficit in (4). The tree and lengths in (3) satisfy every premise of the proposed universal inequality, so this single example disproves that assertion. □

Verification material. The accompanying script `code/exact_three_leaf.py` independently sums over all 20^4 assignments to the four non-root vertices and applies the literal Fitch recursion. From the package root, run `python3 code/exact_three_leaf.py` with Python 3.10 or newer; only the standard library is needed. The displayed finite calculation is the proof; the script provides a reproducible check.

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References

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